24 · Waves: light and sound

The words of this track, as it says them. Bold brown is the examiner.

Waves of light and sound. Four marks.

Explain amplitude, wavelength and frequency, and how we experience each one. Compare light and sound waves, and explain the difference between reflection and refraction of light.

The hook: amplitude is how tall, so loud or bright. Wavelength is peak to peak, so colour. Frequency is how many per second, so pitch. Reflection bounces, refraction bends.

Amplitude is the height of the wave: how far it moves from rest. In sound we experience it as loudness, measured in decibels. In light, as brightness.

Wavelength is the distance from one peak to the next. In light, it decides the colour we see. Visible light is measured in nanometres.

Frequency is the number of waves passing per second, measured in hertz. As frequency goes up, wavelength goes down. In sound, we hear frequency as pitch. Humans hear about twenty to twenty thousand hertz.

Light versus sound. Light is electromagnetic, and needs no medium: it can cross a vacuum. Sound is a mechanical pressure wave, made of compressions and rarefactions, and it needs a medium: a gas, a liquid, or a solid.

Reflection: light bounces back off a flat, opaque surface, like a mirror. A coloured object reflects only its own wavelengths, and absorbs the rest.

Refraction: light bends when it passes into a medium of a different density, because it slows down there. A rainbow in raindrops. And in your own eye, the cornea and the lens refract light to focus it on the retina.

Her checklist says reflection and refraction of sound, too, and her keys cover it. Sound reflects: an echo is reflected sound, and so is sonar. And sound refracts: it bends upward, away from hot ground, because it travels faster in hot air than in cold air. While we're on rainbows, her model answer: raindrops act as tiny prisms, and each wavelength of light slows by a different amount, so blue, the shortest, bends the most, and red the least.

The traps. Loudness is amplitude, and pitch is frequency. Her quizzes swap them to catch you. Second: a higher frequency means a shorter wavelength. Third: sound can't travel through a vacuum. Light can.

Once more. Waves.

Amplitude is the wave's height: loudness in decibels, or brightness. Wavelength is peak to peak: the colour of light. Frequency is waves per second, in hertz: the pitch of sound, and a higher frequency means a shorter wavelength. Light is electromagnetic and needs no

medium; sound is a mechanical pressure wave that needs one. Reflection: light bounces back, like a mirror. Refraction: light bends as it slows in a different density, like a rainbow, or the cornea and lens focusing light. Sound reflects as an echo or sonar, and refracts, bending away from hot ground.